

Data Structures Demo Syllabus

A stack is a linear data structure following last in first out order. Push inserts element and pop removes the most recent element.

Queue operations follow first in first out order and provide a useful comparison to stack behavior in exam answers.

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Binary tree traversal visits nodes in a defined sequence. Preorder processes root-left-right, inorder processes left-root-right, and postorder processes left-right-root.

Figure 2 presents the binary tree traversal structure and the visit order used recursive algorithms.

Example: inorder traversal of a binary search tree lists the keys in sorted order.

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Breadth first search explores a tree level by level, while depth first search follows one branch until backtracking occurs.

The syllabus revision notes relate traversal algorithms to recursion, queue usage, and tree hierarchy diagrams.